

Relay

Extended Literature Survey

15 Additional Papers on Efficient, Robust and Heterogeneous Multi-Agent Communication

Scope. These reviews cover the 15 additional papers selected after the first Relay literature pass. The five papers already reviewed in depth - RIAL/DIAL, CommNet, TarMAC, IC3Net and MAPPO - are intentionally not repeated here.

Review structure

Base / research question

Approach / method

Key findings and conclusions

Gaps / limitations

Implications for Relay

Relay experiments inspired by the paper

Prepared as a working research reference. Findings are paraphrased from the primary papers; links are included with each review.

Executive synthesis

What these 15 add beyond the foundational five. The newer literature moves the problem from simply "can agents learn to communicate?" toward a much more operational set of questions: what information survives a bottleneck, when transmission is worth its cost, which teammate should receive it, how much capacity a message deserves, how receivers use uncertain or stale information, and how learned protocols behave under discrete/noisy/heterogeneous conditions.

1. Heterogeneity is structural, not cosmetic. HetNet is the clearest evidence that class-specific observations, actions and sender-receiver relationships deserve explicit treatment. For Relay, Scout/Ambulance/Firefighter should not be assumed to require symmetric policies or symmetric message semantics.

2. "Bandwidth" has several meanings. BVME/DRMAC compress latent dimensions; IMAC constrains information content; Effective Communication Under Limited Bandwidth quantizes and reasons about entropy; DSMS allocates variable message size. Relay should report exactly which resource is constrained instead of using these terms interchangeably.

3. Silence needs a reason. ETCNet, sparse model-based communication and COCOM all show different ways to suppress redundant communication. A free cooperative channel tends to encourage constant talking, so Relay needs a cost, hard budget, novelty criterion or learned value signal.

4. Network impairment is multidimensional. TMC addresses burst loss and temporal reuse; CoDe and VIL2C address delay/staleness; discrete-channel learning addresses corruption of digital messages. Relay should test loss, latency and corruption separately before combining them.

5. Receiver-side reasoning matters. M2I2, DRMAC, COCOM and T2MAC emphasize that communication quality is not determined only by the sender. The receiver can infer missing information, suppress redundant dimensions, estimate what is already common knowledge, and weigh uncertainty/timeliness.

Recommended Relay experiment progression

1. Unrestricted continuous communication baseline on the clean channel.
2. Fixed message-capacity sweep (latent dimensions), then learned bottleneck/compression.
3. Discrete/quantized messages with explicit vocabulary or estimated bit cost.
4. Communication cost or hard budget; learn when to transmit.
5. Targeted recipient selection and role-pair-specific communication.
6. Variable message size under a shared team bandwidth budget.
7. iid packet loss, then burst loss + receiver caching/TTL.
8. Fixed and stochastic communication latency + message-age awareness.
9. Combined noise/loss/latency and range/topology restrictions.
10. Generalization across map seeds and changed heterogeneous team compositions.

Coverage matrix

#	Paper	Hetero	BW/size	When	Whom	Delay/loss	Discrete
1	HetNet	Y	Y		Y	Y	Y
2	VIL2C			Y	Y	Y	
3	BVME		Y				
4	M2I2			Y			

#	Paper	Hetero	BW/size	When	Whom	Delay/loss	Discrete
5	CoDe					Y	
6	Sparse/MBC	Y		Y			
7	DRMAC		Y				
8	COCOM			Y			
9	T2MAC			Y	Y		
10	DSMS		Y	Y			
11	Limited BW		Y				Y
12	IMAC		Y	Y	Y		
13	ETCNet		Y	Y			
14	TMC			Y		Y	
15	Discrete noise					Y	Y

Y = a direct focus of the paper. Blank = not a primary focus; it may still be compatible with that dimension.

Paper index

1. **Heterogeneous Policy Networks for Composite Robot Team Communication and Coordination (HetNet)** (2024 (IEEE T-RO; extended arXiv version 2026))
2. **VIL2C: Value-of-Information Aware Low-Latency Communication for Multi-Agent Reinforcement Learning** (2026)
3. **Bandwidth-Constrained Variational Message Encoding for Cooperative Multi-Agent Reinforcement Learning (BVME)** (2026)
4. **M2I2: Learning Efficient Multi-Agent Communication via Masked State Modeling and Intention Inference** (2026)
5. **CoDe: Communication Delay-Tolerant Multi-Agent Collaboration via Dual Alignment of Intent and Timeliness** (2025)
6. **Sparse Communication in Multi-Agent Deep Reinforcement Learning** (2025)
7. **Revisiting Communication Efficiency in Multi-Agent Reinforcement Learning from the Dimensional Analysis Perspective (DRMAC)** (2025)
8. **Efficient Communication in Multi-Agent Reinforcement Learning with Implicit Consensus Generation (COCOM)** (2025)
9. **T2MAC: Targeted and Trusted Multi-Agent Communication through Selective Engagement and Evidence-Driven Integration** (2024)
10. **Dynamic Size Message Scheduling for Multi-Agent Communication under Limited Bandwidth (DSMS)** (2024)
11. **Effective Multi-Agent Communication Under Limited Bandwidth** (2024)
12. **Learning Efficient Multi-Agent Communication: An Information Bottleneck Approach (IMAC)** (2020)
13. **Event-Triggered Communication Network With Limited-Bandwidth Constraint for Multi-Agent Reinforcement Learning (ETCNet)** (2021)
14. **Succinct and Robust Multi-Agent Communication With Temporal Message Control (TMC)** (2020)
15. **Communication Learning via Backpropagation in Discrete Channels with Unknown Noise** (2020)

1. Heterogeneous Policy Networks for Composite Robot Team Communication and Coordination (HetNet)

Authors: Esmaeil Seraj, Rohan Paleja, Luis Pimentel, Kin Man Lee, Zheyuan Wang, Daniel Martin, Matthew Sklar, John Zhang, Zahi Kakish, Matthew Gombolay

Year / venue: 2024 (IEEE T-RO; extended arXiv version 2026) - IEEE Transactions on Robotics

Primary link: <https://arxiv.org/abs/2606.20962>

Base / research question

Relay is intrinsically heterogeneous: a Scout, Ambulance and Firefighter do not merely occupy different positions; they have different capabilities, observations, action spaces and reasons to communicate. HetNet starts from almost exactly that class of problem. Rather than assuming a homogeneous swarm that can share one policy and one generic message encoder, the paper models a team as multiple robot classes and asks how class-specific policies and communication should be learned jointly.

The central idea is a heterogeneous communication graph. The meaning and usefulness of a message can depend on both the sender class and the receiver class, so the architecture learns class-aware processing instead of forcing every robot through the same communication pathway. The extended work also pushes the learned messages toward compact, binarized representations and studies transfer to different team compositions and channel conditions.

Approach / method

The policy is organized as a heterogeneous graph-attention network. Nodes correspond to agents, but class information determines how observations are encoded, how messages are generated and how received information is fused. Sender-receiver relationships are therefore not treated as interchangeable. This matters in a composite team because information from a perception-heavy agent may have a different semantic role for a mobility or manipulation agent.

Training uses a PPO-family method adapted to heterogeneous teams (MAH-PPO). Communication is learned end-to-end with the task objective rather than supervised with pre-labelled message meanings. The paper also studies low-bandwidth communication by binarizing message representations and evaluates robustness under noisy/lossy channels. A Robotarium FireCommander demonstration connects the framework to a physical multi-robot search-and-response style task.

Key findings / conclusions

The paper reports large gains over the compared homogeneous or less specialized baselines across several heterogeneous-team domains, with the exact gain depending strongly on the task. The extended experiments also show that compact binary communication can preserve most of the cooperative benefit while dramatically reducing the representation size, and that the learned system degrades gracefully under channel corruption rather than immediately collapsing.

An equally important qualitative conclusion is that heterogeneity is not a nuisance to be hidden by parameter sharing. The class structure can be exploited: agents can learn different internal representations and different communication patterns because their informational roles are different. The physical demonstration reinforces that the architecture is at least compatible with the kind of perception-to-action information flow that Relay ultimately wants.

Gaps / limitations

The robot classes are known in advance, so the method does not discover roles. That is actually compatible with Relay, but it means the paper does not answer whether role boundaries themselves should emerge.

The physical demonstration is not the same as end-to-end learning on hardware; most learning and robustness analysis remain simulation-centric. The heterogeneous graph stack is also substantially more complex than a minimal MAPPO baseline, so reproducing the whole architecture too early could obscure whether gains come from communication or from extra model capacity.

The 2024 journal publication and the 2026 extended arXiv version should be distinguished when citing results, because the extended version contains additional experiments and bandwidth/noise analysis.

Implications for Relay

This is the strongest architectural evidence in the set for keeping role-specific policy/encoder heads instead of forcing Scout, Ambulance and Firefighter into one identical observation-to-message pipeline. Relay can still share low-level components where useful, but communication should be allowed to be asymmetric across role pairs.

It also suggests an experiment that is much more interesting than a single global communication channel: compare one shared message encoder/decoder against role-pair-specific encoders/decoders. If a Scout message intended for an Ambulance differs systematically from one intended for a Firefighter, that is evidence of meaningful specialization rather than generic latent-state broadcasting.

Finally, the binarized-message and channel-noise experiments provide a bridge from a convenient differentiable training representation to deployable messages. Relay does not need to implement the full HetNet stack initially, but this paper should inform the heterogeneous-policy design and the later binary/noisy-channel stage.

Relay experiments inspired by this paper

- Shared communication module vs role-specific sender/receiver modules.
- Train on one team composition and test changed counts of Scout/Ambulance/Firefighter agents.
- Continuous message vector vs binarized message representation at matched capacity.
- Role-pair communication ablation: Scout->Ambulance, Scout->Firefighter, responder->Scout, etc.
- Noise/loss sweep after clean-channel training and after robustness-aware training.

Bottom line: Highest-priority paper for Relay's heterogeneous-role architecture and a realistic bridge from learned communication to multi-robot deployment.

2. VIL2C: Value-of-Information Aware Low-Latency Communication for Multi-Agent Reinforcement Learning

Authors: Qian Zhang, Zhuo Sun, Yao Zhang, Zhiwen Yu, Bin Guo, Jun Zhang

Year / venue: 2026 - AAAI 2026

Primary link: <https://arxiv.org/abs/2511.19146>

Base / research question

Most Comm-MARL papers act as if a message generated at time t is available to teammates immediately. VIL2C attacks that assumption directly. It asks what happens when wireless communication consumes time and the message may already be stale by the time it arrives. That makes latency part of the coordination problem rather than an afterthought.

The key concept is value of information (VoI): a message is worth transmitting quickly when it can materially change the receiver's action decision. Communication resources should therefore be prioritized according to decision impact, not merely message age or sender identity.

Approach / method

VIL2C extends the decentralized partially observable decision process with explicit communication latency. It estimates the decision value of a message from the change it induces in the receiver's action distribution, using a KL-divergence style measure, and normalizes this by latency. Messages with higher value per unit latency receive higher priority.

The framework then allocates communication resources such as bandwidth/power to important messages. It also introduces progressive reception: the receiver can stop waiting for additional information once its action distribution is confident enough, reducing unnecessary delay. The learning backbone is compatible with CTDE and is demonstrated with MAPPO-style training across cooperative benchmarks.

Key findings / conclusions

Across the evaluated tasks and communication-resource settings, the method reduces communication-induced decision latency while preserving cooperative performance substantially better than latency-oblivious alternatives. The results support the intuitive but important claim that not every message deserves equal urgency.

The paper also shows that latency and information value interact: the same payload can be useful when fresh and nearly irrelevant when stale. That means a realistic communication policy should reason about message age/timeliness, not just whether a packet arrives.

Gaps / limitations

The full system is networking-heavy: channel state, power/bandwidth allocation, acknowledgements and entropy-based stopping rules add assumptions that Relay does not need for its first research milestone.

VoI depends on estimating how messages alter action distributions. That is elegant but adds inference and calibration overhead, and the chosen thresholds/resource model may be domain-specific.

The work primarily optimizes communication delivery around an existing learned policy; it is less focused on interpreting the semantics that emerge inside the messages.

Implications for Relay

Relay should explicitly distinguish packet loss from latency. A delayed victim-location message is not equivalent to a dropped one: it may arrive, but too late to be useful. Fire-state information may be even more time-sensitive because hazards can evolve.

For an initial Relay experiment, the full wireless optimizer is unnecessary. A much cleaner adaptation is to attach an age/timestamp to each received message and test fixed delay, uniformly random delay, and bursty/time-varying delay. Performance can then be plotted against mean and tail latency.

Later, Relay can test a simpler VoI-inspired scheduler: if only one of several pending messages can be delivered, prioritize the message whose removal most changes the receiver's action distribution or critic estimate.

Relay experiments inspired by this paper

- 0/1/2/4/8-step fixed communication delay.
- Random and bursty delay distributions with the same mean latency.
- Fresh-message policy vs cached stale-message policy.
- FIFO scheduling vs age-based vs VoI-inspired scheduling under a shared channel budget.
- Measure task success and decision changes as a function of message age.

Bottom line: The key modern reference for treating latency and staleness as first-class Relay communication variables.

3. Bandwidth-Constrained Variational Message Encoding for Cooperative Multi-Agent Reinforcement Learning (BVME)

Authors: Wei Duan, Jie Lu, En Yu, Junyu Xuan

Year / venue: 2026 - AAMAS 2026

Primary link: <https://arxiv.org/abs/2512.11179>

Base / research question

BVME focuses on a common weakness in learned communication: agents often transmit high-dimensional continuous embeddings simply because the network is allowed to. The paper asks whether agents can retain cooperative performance when the message representation is forced through a tight information bottleneck.

Instead of hand-selecting dimensions, it uses a variational latent representation so the sender learns a compressed distribution that preserves decision-relevant information while discarding unnecessary detail.

Approach / method

The message encoder produces a diagonal Gaussian posterior. A KL-divergence penalty pulls that posterior toward an uninformative prior, creating a tunable information bottleneck. The method can therefore trade communication capacity against task performance during training rather than truncating a learned vector after the fact.

The approach is applied within cooperative MARL architectures on SMAC/SMACv2 and MPE-style tasks. During evaluation, the mean of the learned posterior can be used deterministically. The paper studies aggressive reductions in message dimensionality and reports modest additional training overhead.

Key findings / conclusions

BVME reports that substantial dimensional compression - roughly two-thirds to more than four-fifths fewer message dimensions in the studied settings - can retain comparable or sometimes better cooperative performance. This supports the hypothesis that unconstrained communication vectors contain large amounts of redundant information.

The method is especially attractive under severe capacity constraints and sparse communication structures, where simply shrinking a standard message vector tends to degrade more sharply.

Gaps / limitations

A reduction in continuous latent dimensions is not automatically a reduction in transmitted bits. A 4-dimensional float vector can still be expensive compared with a 4-bit symbol. BVME therefore solves representation compression more directly than physical-link accounting.

The Gaussian variational formulation still uses continuous values at execution unless a later quantization step is added. The paper also builds on specific graph/value-learning setups that are not identical to Relay's planned MAPPO baseline.

Because the work is very recent, there is less independent replication than for older bottleneck methods such as IMAC.

Implications for Relay

Relay can borrow the principle without adopting the whole architecture: learn a high-capacity communication policy, then train variants with explicit bottleneck strengths and compare the reward-versus-capacity frontier.

This provides a better experiment than only choosing arbitrary message dimensions. If a 4-D message works, the important question is whether it works because four independent quantities are necessary or because the network is using only a fraction of the available entropy.

BVME is therefore a strong companion to discrete/entropy papers: use it for continuous bottleneck experiments, then quantize the best compact representation to test actual symbol/bit budgets.

Relay experiments inspired by this paper

- Message dimension sweep with and without variational bottleneck.
- KL-weight sweep and performance/capacity Pareto curve.
- Naive truncation vs learned bottleneck at equal dimension.
- Continuous bottleneck vs later quantized codebook at comparable effective information.
- Inspect whether different roles occupy different parts of the compressed latent space.

Bottom line: Best recent reference for learned compression under hard representation constraints, but should not be confused with true bit-level bandwidth.

4. M2I2: Learning Efficient Multi-Agent Communication via Masked State Modeling and Intention Inference

Authors: Chuxiong Sun, Peng He, Qirui Ji, Zehua Zang, Jiangmeng Li, Rui Wang, Wei Wang

Year / venue: 2026 - AAAI 2026

Primary link: <https://arxiv.org/abs/2501.00312>

Base / research question

M2I2 shifts attention from only deciding what a sender should transmit to what a receiver can reconstruct or infer without transmission. Its premise is that teammates do not need every raw feature if they can infer missing state and intention from context.

That makes communication efficiency partly a representation-learning problem: the system should learn which state dimensions are worth exposing, which can be masked, and what a teammate is likely to do next.

Approach / method

The method combines masked-state modeling with intention inference. A masked-autoencoder-style objective trains representations that reconstruct hidden state information, while an inverse/joint-action component learns signals about teammate intentions. A dimensional rational network learns which observation dimensions deserve emphasis or masking.

The authors integrate these ideas with several MARL backbones, including value-decomposition methods and actor-critic methods such as MAPPO/MADDPG, to show that the communication layer is not tied to one learner.

Key findings / conclusions

The reported results show improved communication efficiency and task performance across the tested cooperative settings. Visualizations indicate that the learned importance of dimensions changes across agents, entity types and timesteps rather than behaving like a fixed feature selector.

The broader conclusion is that useful communication is relational and contextual: what one teammate needs at one moment may be predictable or irrelevant at another.

Gaps / limitations

M2I2 adds several auxiliary objectives and learned modules. That increases implementation complexity and creates more places where gains can come from representation learning rather than the communication mechanism itself.

It does not directly model real packet size, discrete symbols, latency or channel noise. A masked latent dimension is still an abstract neural representation.

Reconstructing hidden state is useful only when the environment has enough temporal/relational structure for the missing information to be predictable.

Implications for Relay

The most useful lesson for Relay is receiver-side inference. If the Ambulance can infer the Scout's likely motion or already knows a mapped corridor, the Scout should not repeatedly retransmit that information.

The discovered-map representation is a natural test bed. Relay can mask portions/features of communicated map state and ask whether the receiver can infer enough context to act correctly. This also suggests measuring which semantic variables survive learned compression.

M2I2 is better treated as an advanced robustness/efficiency experiment after the base communication pipeline works, rather than a starting architecture.

Relay experiments inspired by this paper

- Feature/dimension masking on message inputs and received embeddings.
- Receiver reconstruction of omitted map/object information.
- Auxiliary intention prediction vs no intention prediction.
- Measure whether message importance differs by role and task phase.
- Test whether inference reduces communication without harming rescue success.

Bottom line: Strong recent paper for receiver-side inference and learned feature selection; valuable later, but too elaborate for Relay's first baseline.

5. CoDe: Communication Delay-Tolerant Multi-Agent Collaboration via Dual Alignment of Intent and Timeliness

Authors: Shoucheng Song, Youfang Lin, Sheng Han, Chang Yao, Hao Wu, Shuo Wang, Kai Lv

Year / venue: 2025 - AAAI 2025

Primary link: <https://arxiv.org/abs/2501.05207>

Base / research question

CoDe starts from asynchronous communication: messages can arrive after the world and sender have already changed. It therefore asks how a receiver should interpret a delayed message rather than treating every received packet as current truth.

The paper argues that robust delayed communication needs two forms of alignment: semantic alignment with the sender's likely intent and temporal alignment with the age/timeliness of the message.

Approach / method

The sender learns an intent representation based partly on inferred future actions. The receiver then integrates incoming communication with a dual-alignment mechanism that evaluates both the relevance of the transmitted intent and how timely the message still is.

Experiments cover fixed and time-varying delays across cooperative benchmarks such as SMAC, GRF and Hallway. The objective is not merely to tolerate missing packets but to use delayed packets intelligently when they still contain useful predictive information.

Key findings / conclusions

CoDe improves robustness under communication delays compared with delay-unaware alternatives and maintains stronger coordination as delays increase or vary. The results make a useful distinction between payload content and temporal validity: a semantically good message can still be harmful if treated as fresh after a long delay.

The authors also acknowledge that predicting useful future intent has a limited horizon. Under severe delays, the sender's forecast becomes less reliable and the advantage decreases.

Gaps / limitations

Future-action/intent prediction can itself become inaccurate in highly stochastic environments or when teammates react strongly to one another, producing a moving target.

The method handles delay more directly than bandwidth or discrete channel capacity, so it does not replace a compression/gating method.

The architecture is more complex than simply attaching message age and allowing the receiver network to learn a decay/attention function.

Implications for Relay

Relay has an especially clear semantic split between persistent and transient information. A discovered victim location may remain valid for a long time, while a robot trajectory or spreading-fire state may become stale quickly. CoDe suggests that message age should interact with message type/content.

A clean Relay adaptation is to include message age in the receiver observation and compare this against a receiver that is blind to age. A later version can learn intent/future-state embeddings for dynamic information.

This paper should directly shape the delayed-communication experiment design even if the full dual-alignment architecture is not implemented.

Relay experiments inspired by this paper

- Receiver with vs without explicit message-age input.
- Persistent victim/location messages vs transient hazard/agent-state messages under equal delay.
- Fixed, random and bursty delay.
- Age-based exponential decay vs learned timeliness attention.
- Intent/future-action message vs current-state message under increasing delay.

Bottom line: Core reference for stale-message semantics and delay-tolerant coordination.

6. Sparse Communication in Multi-Agent Deep Reinforcement Learning

Authors: Shuai Han, Mehdi Dastani, Shihan Wang

Year / venue: 2025 - Neurocomputing

Primary link: <https://www.sciencedirect.com/science/article/pii/S0925231225000165>

Base / research question

This paper tackles communication redundancy from a predictive angle. Instead of asking only whether a sender thinks its message is important, it asks whether teammates could already predict that message. If the answer is yes, transmission is unnecessary.

This is attractive for cooperative teams because many observations change slowly or become common knowledge. Repeatedly transmitting predictable information wastes bandwidth without adding coordination value.

Approach / method

The proposed model-based communication mechanism trains message estimators that predict what another agent would send. A transmission is triggered when the actual intended message differs enough from what the receiver is expected to infer. Communication scheduling is decentralized rather than relying on a central runtime controller.

The journal study evaluates both homogeneous and heterogeneous multi-agent domains and measures task quality alongside communication overhead.

Key findings / conclusions

The method reduces communication substantially while maintaining or improving cooperative performance in the evaluated tasks. The heterogeneous results are important because predictability does not require every agent to have identical policies or observations.

The key conceptual result is that novelty can be defined relative to the receiver's predictive model. A message is valuable when it carries information the teammate cannot already reconstruct.

Gaps / limitations

The estimator itself requires training and compute, and a poor estimator can suppress messages that were actually important. The threshold for deciding prediction error is therefore consequential.

Rapidly changing or nonstationary policies can make message prediction unreliable. Communication may also become overly sparse if the sender and receiver disagree about what is already known.

The paper mainly reduces transmission frequency; it does not by itself solve how many bits each transmitted message should contain.

Implications for Relay

This maps naturally onto Relay's Scout. Repeating "corridor clear" every timestep is wasteful; reporting a newly discovered victim, blocked passage or fire is novel and should pass the gate.

The idea also gives Relay a more defensible communication-efficiency objective than arbitrary periodic sending. The agent communicates on informational surprise rather than on a fixed timer.

A simpler first version can avoid training a separate message predictor: compare current candidate message to the last transmitted message or to a receiver belief estimate, then later adopt model-based prediction.

Relay experiments inspired by this paper

- Periodic communication vs prediction-error/novelty-triggered communication.
- Threshold sweep: communication rate vs rescue performance.
- Compare homogeneous-style shared predictor with role-pair-specific predictors.
- Inject sudden map/hazard changes and measure whether the gate reactivates quickly.
- Combine sparse scheduling with a fixed discrete message budget.

Bottom line: One of the best modern references for deciding when not to communicate because teammates can already infer the content.

7. Revisiting Communication Efficiency in Multi-Agent Reinforcement Learning from the Dimensional Analysis Perspective (DRMAC)

Authors: Chuxiong Sun, Peng He, Rui Wang, Changwen Zheng

Year / venue: 2025 - AAMAS 2025

Primary link: <https://arxiv.org/abs/2501.02888>

Base / research question

DRMAC argues that existing communication-efficiency work often stops too early. A sender may choose the right timing and suppress some messages, yet the received latent embedding can still contain redundant or confounding dimensions. The paper therefore examines efficiency inside the message representation itself.

The research question is receiver-centric: after messages have been generated and aggregated, which dimensions actually influence the decision, and can irrelevant dimensions be suppressed during learning?

Approach / method

The method adds redundancy-reduction objectives and a dimensional masking mechanism. Gradients and representation learning are shaped so that decision-relevant dimensions are emphasized and redundant dimensions are attenuated.

The authors demonstrate the approach as a plug-in across multiple MARL and communication backbones, including MAPPO and established communication methods, making the idea less architecture-specific than a standalone end-to-end system.

Key findings / conclusions

The experiments report stronger communication efficiency and cooperative performance when redundant dimensions are explicitly controlled. The paper reinforces a useful distinction: reducing the number of transmissions and reducing the information carried inside each transmission are separate problems.

It also suggests that receiver-side dimensional analysis can expose communication inefficiency that sender-side gates do not see.

Gaps / limitations

Latent dimensional redundancy is still not equivalent to actual wireless bandwidth. A single active float dimension can cost more than several discrete symbols depending on encoding.

The regularization/masking adds another training objective and can make interpretation sensitive to representation rotation: neural latent dimensions are not guaranteed to correspond to stable human-readable factors.

The method does not directly solve latency, loss or recipient selection.

Implications for Relay

Relay should not report only "message dimension = 8" as a communication result. It should ask how many dimensions/symbols are actually necessary and whether deleting individual dimensions changes behavior.

DRMAC motivates post-training dimension ablations, learned masks and effective-rank/entropy analysis of the message embedding. This is useful for demonstrating that an emergent protocol is compact rather than merely living in a large vector.

Because the mechanism is compatible with MAPPO-style training, it is a plausible later ablation without replacing Relay's main learner.

Relay experiments inspired by this paper

- Post-training single-dimension and group ablations.
- Learned dimensional mask vs fixed smaller message vector.
- Effective rank / variance / entropy analysis of role-specific messages.
- Communication frequency reduction alone vs dimensional reduction alone vs both.
- Receiver-side masking under packet loss to test redundancy robustness.

Bottom line: Important for measuring and removing hidden redundancy inside learned messages, not just counting how often agents talk.

8. Efficient Communication in Multi-Agent Reinforcement Learning with Implicit Consensus Generation (COCOM)

Authors: Dapeng Li, Na Lou, Zhiwei Xu, Bin Zhang, Guoliang Fan

Year / venue: 2025 - AAAI 2025

Primary link: <https://ojs.aaai.org/index.php/AAAI/article/view/34490>

Base / research question

COCOM is built around a simple insight: communication should add information beyond what teammates can already agree on from their local context. It combines an implicit shared consensus with explicit messages that are reserved for residual information.

This reframes efficient communication as "what does my teammate not already know?" rather than "what do I know?" - a particularly useful distinction for teams that gradually build a common map or task belief.

Approach / method

During CTDE, the framework learns local consensus representations guided by a global consensus signal. Communication is encouraged to carry information that complements this implicit consensus. A gating mechanism controls whether explicit communication occurs, and cross-attention integrates received information.

Mutual-information-related objectives are used to make messages informative relative to the consensus representation. The method is evaluated with multiple cooperative backbones and benchmark tasks.

Key findings / conclusions

The paper reports reduced communication overhead while retaining strong coordination performance. The results support the idea that explicit communication and implicit shared belief can substitute for one another to some extent.

The authors explicitly note a current limitation: the communication is broadcast rather than individually customized for each teammate, leaving targeted residual communication as future work.

Gaps / limitations

Broadcast communication is less suitable for strongly heterogeneous teams where Ambulance and Firefighter need different information.

The consensus representation learned during CTDE may be hard to interpret, and the mutual-information machinery adds optimization complexity.

The work does not directly model latency, discrete bit budgets or real channel errors.

Implications for Relay

Relay's discovered map makes the consensus idea concrete. Once all agents know a corridor or victim location, retransmitting it should be unnecessary; messages should focus on newly discovered or role-relevant deltas.

The paper suggests maintaining an explicit shared/belief state for analysis even if the policy does not receive the global map directly. We can compare messages against what the receiver has already discovered and quantify novelty.

Because COCOM broadcasts, Relay can extend the idea with role-targeted residual messages: "new fire here" should go to Firefighter, while "victim here" should preferentially go to Ambulance.

Relay experiments inspired by this paper

- Full-state/broadcast message vs residual/delta message.

- Communication rate as common map knowledge increases over an episode.
- Novel-information gate based on receiver belief overlap.
- Broadcast residual message vs role-targeted residual message.
- No-communication fallback after consensus has formed.

Bottom line: Useful conceptual bridge between shared world knowledge and explicit messages; especially relevant to Relay's discovered-map mechanics.

9. T2MAC: Targeted and Trusted Multi-Agent Communication through Selective Engagement and Evidence-Driven Integration

Authors: Chuxiong Sun, Zehua Zang, Jiabao Li, Jiangmeng Li, Xiao Xu, Rui Wang, Changwen Zheng

Year / venue: 2024 - AAAI 2024

Primary link: <https://arxiv.org/abs/2401.10973>

Base / research question

T2MAC revisits three questions together: when should an agent communicate, whom should it engage, and how much should the receiver trust what it receives? This is a modern targeted-communication treatment rather than a pure broadcast scheme.

The trust component is important because a message can be uncertain even when the channel is perfect. Partial observations and ambiguous local evidence can make the sender itself unsure.

Approach / method

The architecture constructs individualized information for teammates and uses selective engagement to choose communication windows and partners. At the receiver, an evidence-based representation produces uncertainty/evidence signals that influence how messages are fused into the policy.

The method is evaluated across cooperative MARL settings and is shown with multiple learning backbones, including MAPPO-style actor-critic training.

Key findings / conclusions

The paper reports a favorable balance between task performance and communication rate compared with its evaluated baselines. The main qualitative finding is that selective recipient engagement and uncertainty-aware integration can prevent both unnecessary transmissions and harmful over-reliance on weak evidence.

It extends the targeted-communication story beyond "address a teammate" by making the receiver decide how much confidence to place in the received information.

Gaps / limitations

The communication remains an abstract neural representation rather than a bit-accurate physical channel. Explicit latency and packet corruption are outside the central formulation.

Evidence/uncertainty calibration can be fragile, especially outside the training distribution. A low-confidence message is not necessarily wrong, and an overconfident sender can still mislead.

The full architecture introduces more machinery than necessary to test simple role-targeted communication.

Implications for Relay

T2MAC is directly relevant to Relay because recipients have different responsibilities. The Scout should not need to send every discovery to every responder. Recipient selection itself can be learned and evaluated.

Uncertainty is also meaningful in Relay even without perception-classification noise: a Scout may have incomplete map knowledge, an old hazard observation, or uncertain reachability. The message can carry or induce confidence and the receiver can condition its response on it.

A clean first experiment is targeted vs broadcast communication; uncertainty-aware fusion can be a later extension.

Relay experiments inspired by this paper

- Broadcast to all vs learned recipient selection.

- Role-targeted heuristic routing vs learned targeting.
- Attach confidence/age to messages and test uncertainty-aware fusion.
- Measure wasted transmissions to agents whose role cannot act on the information.
- Ablate targeting and trust modules independently.

Bottom line: Best recent companion to TarMAC-style targeting for Relay's role-specific recipient selection.

10. Dynamic Size Message Scheduling for Multi-Agent Communication under Limited Bandwidth (DSMS)

Authors: Qingshuang Sun, Denis Steckelmacher, Yuan Yao, Ann Nowé, Raphaël Avalos

Year / venue: 2024 - IEEE Transactions on Mobile Computing

Primary link: <https://arxiv.org/abs/2306.10134>

Base / research question

Many communication schedulers make a binary choice: send a fixed-size message or stay silent. DSMS argues that this wastes a crucial degree of freedom. Under a shared bandwidth budget, a very important observation may deserve a larger message while routine information may deserve only a few bytes or nothing at all.

The problem is therefore joint scheduling and sizing: decide not only who talks, but how much of the shared channel each message receives.

Approach / method

DSMS learns an importance-dependent bandwidth allocation. Candidate message vectors are compressed to the allocated size using a clipped Fourier-transform representation, then reconstructed by the receiver. The communication scheduler operates under an overall channel constraint rather than giving every sender a fixed payload size.

The method is trained in CTDE settings and evaluated in partially observable cooperative navigation/predator-prey style tasks. Analyses relate allocated message size to situational variables such as distance, roles and novelty.

Key findings / conclusions

Dynamic message sizing improves the performance-bandwidth tradeoff over fixed-size schemes in the tested environments. The allocation patterns also show that message length can become context-sensitive rather than constant across agents and timesteps.

The broader conclusion is useful for Relay: communication capacity itself can be an action. Silence, one-symbol updates and larger high-priority messages can coexist within one learned protocol.

Gaps / limitations

The Fourier compression mechanism assumes useful structure in the latent vector and is not an obvious fit for discrete emergent symbols.

The benchmark set is relatively small, and byte-level allocation in simulation still abstracts away packet headers, retransmission and other networking overhead.

Dynamic sizing can make training and evaluation harder to interpret because communication frequency and payload size change simultaneously.

Implications for Relay

Relay should eventually separate two budgets: number of transmissions and total transmitted symbols/bits. Two methods with the same number of messages can have radically different bandwidth cost.

A simpler Relay version can avoid Fourier transforms: let the communication action choose among payload sizes such as 0, 1, 2, 4 or 8 symbols, with a shared per-timestep budget.

This would be especially interesting when rare rescue-critical discoveries deserve richer messages while routine navigation requires little or no communication.

Relay experiments inspired by this paper

- Fixed-size vs variable-size messages under equal total symbol budget.
- Per-timestep shared team bandwidth budget.
- Discrete size choices: 0/1/2/4/8 symbols.
- Analyze which events trigger large messages.
- Frequency-only cost vs total-symbol cost.

Bottom line: Key paper for moving Relay beyond binary send/silence toward variable message size under a shared bandwidth budget.

11. Effective Multi-Agent Communication Under Limited Bandwidth

Authors: Lebin Yu, Qiexiang Wang, Yunbo Qiu, Jian Wang, Xudong Zhang, Zhu Han

Year / venue: 2024 - IEEE Transactions on Mobile Computing

Primary link: <https://ieeexplore.ieee.org/document/10342771/>

Base / research question

This work is valuable because it treats bandwidth more literally than many latent-dimension papers. It asks how learned messages can be quantized and how the discrete entropy of those messages can be constrained so communication cost corresponds more closely to actual coded information.

The core tension is explicit: maximize cooperative return while satisfying a limited-bandwidth constraint rather than merely adding a vague regularizer and hoping messages become smaller.

Approach / method

The framework quantizes learned messages and formulates communication cost through discrete entropy. Because direct gradients through discrete entropy/quantization are problematic, the paper introduces a surrogate entropy minimization mechanism and a soft-barrier optimization method for the constrained objective.

The experiments evaluate multiple environment settings under several bandwidth limits, testing whether performance can be preserved as the available communication rate is tightened.

Key findings / conclusions

The results show that learned quantization and entropy control can achieve substantially better cooperation-bandwidth tradeoffs than unconstrained or less targeted compression approaches in the studied settings.

The paper's main conceptual contribution for Relay is the distinction between representation size and information rate. A neural vector with few dimensions is not necessarily low-bandwidth; a discrete distribution with low entropy is much closer to something that can be assigned a bit cost.

Gaps / limitations

Entropy is still an idealized coding measure. Actual packets have headers, synchronization and channel-coding overhead that are not captured by the policy-level objective.

The method focuses on message content/cost more than on who should communicate or when, so it should be paired with scheduling/targeting mechanisms.

Quantization and constrained optimization can make training less stable than a continuous-message baseline.

Implications for Relay

If Relay eventually claims "N-bit communication," this is one of the papers that prevents us from making a sloppy claim based only on latent dimension. We need a discrete alphabet/codebook and an accounting rule for actual symbols or entropy.

A sensible progression is continuous messages first, then categorical/quantized symbols, then an entropy penalty or hard symbol budget. That preserves a clean baseline while still letting the final study talk about meaningful communication constraints.

This paper pairs especially well with IMAC (information bottleneck), DSMS (variable message size), and the discrete-noisy-channel paper.

Relay experiments inspired by this paper

- Continuous float messages vs quantized discrete codebook.

- Vocabulary size sweep (e.g., 2/4/8/16 symbols).
- Uniform symbol use vs entropy-regularized symbol use.
- Hard bit/symbol budget vs soft communication penalty.
- Report reward/success against estimated bits per episode, not only message dimension.

Bottom line: Essential if Relay wants a defensible bit/bandwidth story rather than calling low-dimensional float vectors "low bandwidth."

12. Learning Efficient Multi-Agent Communication: An Information Bottleneck Approach (IMAC)

Authors: Rundong Wang, Xu He, Runsheng Yu, Wei Qiu, Bo An, Zinovi Rabinovich

Year / venue: 2020 - ICML 2020

Primary link: <https://arxiv.org/abs/1911.06992>

Base / research question

IMAC is older than most of this set but remains foundational for bandwidth-aware emergent communication. It formalizes the intuition that a good message should preserve task-relevant information while discarding irrelevant detail, and it jointly considers who communicates and what is sent.

The paper connects limited communication capacity to the entropy/information content of learned messages, making the information-bottleneck principle part of the MARL communication objective.

Approach / method

The communication module contains a protocol for generating messages and a scheduler for selecting communication relationships. A variational information-bottleneck style objective penalizes excessive information while preserving signals useful for cooperative action.

The paper also discusses the relationship between message entropy, quantization and source-coding requirements, providing theoretical motivation for treating entropy as a communication resource rather than just vector dimension.

Key findings / conclusions

Across its benchmark settings and bandwidth constraints, IMAC improves the tradeoff between cooperative performance and communication usage. The experiments support the idea that agents can learn compact messages instead of simply transmitting large hidden states.

Its lasting contribution is conceptual: "what to communicate" and "who should communicate" are coupled under bandwidth constraints. An agent should not spend capacity on information that is either irrelevant or already available to the receiver.

Gaps / limitations

The experiments predate newer benchmarks and modern robustness studies. Latency, burst loss and realistic channel dynamics are not central to the formulation.

Entropy/bottleneck quantities are still proxies for deployable packet cost unless combined with explicit discretization/quantization.

The scheduler and information objective add hyperparameters that can make comparisons to a plain MAPPO communication baseline less transparent.

Implications for Relay

IMAC should provide the theoretical backbone for Relay's compression experiments. We can motivate a message-capacity sweep as an information-bottleneck study rather than an arbitrary architecture choice.

It also suggests measuring both task return and message information content. If performance stays high while entropy falls, the policy has learned a more efficient protocol rather than merely a smaller network.

Relay can implement a simpler bottleneck regularizer first and reserve the full scheduler architecture for comparison if needed.

Relay experiments inspired by this paper

- Information/entropy regularization sweep.
- Performance vs message entropy/capacity.
- Scheduler on/off under equal content bottleneck.
- Receiver-targeted vs broadcast bottlenecked communication.
- Compare compact continuous bottleneck to compact discrete vocabulary.

Bottom line: Still one of the clearest theoretical foundations for Relay's "what information survives a bandwidth limit?" question.

13. Event-Triggered Communication Network With Limited-Bandwidth Constraint for Multi-Agent Reinforcement Learning (ETCNet)

Authors: Guangzheng Hu, Yuanheng Zhu, Dongbin Zhao, Mengchen Zhao, Jianye Hao

Year / venue: 2021 - IEEE Transactions on Neural Networks and Learning Systems

Primary link: <https://arxiv.org/abs/2010.04978>

Base / research question

ETCNet addresses the "when to communicate" problem under a limited channel. Rather than letting every agent exchange a message every timestep, it learns event-triggered decisions about participation in communication.

The important difference from a purely optional gate is that communication is tied to an explicit bandwidth constraint/cost. This gives the policy a reason to remain silent when information is not worth the resource expenditure.

Approach / method

The work develops event-triggered sending and receiving components. A sender-side network decides when a transmission is necessary; a receiver-side component can also decide whether communication should be used. The bandwidth constraint is transformed into an optimization signal/penalty so that communication decisions can be learned jointly with control.

The approach is evaluated in cooperative multi-agent control settings across different communication limitations.

Key findings / conclusions

ETCNet shows that event-triggered communication can maintain strong cooperation while reducing channel use, especially compared with always-on communication under a constrained budget.

For Relay, the deeper result is that silence generally needs an incentive. In fully cooperative tasks, if communication is free and always available, a learned gate may converge toward communicating almost all the time because there is little downside.

Gaps / limitations

The bandwidth model is primarily about whether agents participate in communication rather than the exact number of bits in each message.

The framework is less focused on message semantics and recipient-specific content than later targeted methods.

Thresholds/penalties need careful calibration: too cheap yields always-on behavior; too expensive can suppress critical coordination.

Implications for Relay

Relay should include either a hard communication budget or explicit communication cost when testing emergent gating. Otherwise a result that "agents learned to communicate" may simply mean they learned to leave the channel permanently on.

ETCNet also motivates separating send-side and receive-side choices. A sender may want to talk, but a receiver might decide the information is unnecessary given its own observation.

The minimal Relay version is straightforward: add a binary send action with per-message cost or a team-level quota and measure whether communication concentrates around discoveries, handoffs and emergencies.

Relay experiments inspired by this paper

- Free communication vs explicit per-message penalty.
- Hard messages-per-episode budget vs soft cost.
- Sender gate only vs receiver gate only vs both.
- Plot communication events around victim/fire discovery and rescue phases.
- Budget generalization: train at one limit, test tighter/looser limits.

Bottom line: Core practical reference for ensuring Relay's communication gate has an actual reason to learn silence.

14. Succinct and Robust Multi-Agent Communication With Temporal Message Control (TMC)

Authors: Sai Qian Zhang, Jieyu Lin, Qi Zhang

Year / venue: 2020 - NeurIPS 2020

Primary link: <https://arxiv.org/abs/2010.14391>

Base / research question

TMC exploits temporal redundancy. In many cooperative tasks, an agent's useful message does not change much from one timestep to the next. Re-sending nearly identical content wastes bandwidth, while retaining the last valid message at the receiver can provide robustness when packets are lost.

The paper therefore links communication sparsity and robustness: suppress redundant transmissions in normal operation and use temporal memory to survive lossy channels.

Approach / method

The sender is regularized toward temporally smooth messages and transmits only when the new message differs sufficiently from the most recently sent one, subject to timeout-style safeguards. The receiver stores recent messages and uses validity/expiry logic so that a missing packet does not immediately erase teammate information.

The experiments include communication-loss settings with temporally correlated/bursty loss models, not only independent random packet drops.

Key findings / conclusions

TMC reports large communication reductions - up to roughly 80% in the studied settings - while maintaining task performance, and it remains more robust when messages are lost because receivers can rely on cached information.

The results also show why independent Bernoulli packet loss is not the only useful robustness test. Real link failures can occur in bursts due to occlusion or changing connectivity, making temporal control especially relevant.

Gaps / limitations

The method assumes temporal redundancy. If the world changes rapidly, reusing the last message can be actively harmful.

Choosing similarity thresholds, temporal windows and expiry time introduces task-dependent hyperparameters. Too long a cache can create stale-information failures.

Temporal smoothing may suppress genuinely abrupt but critical updates if the representation or threshold is poorly calibrated.

Implications for Relay

Relay has both slowly changing and rapidly changing information, which makes TMC unusually useful. Static victim locations or mapped corridors can be cached safely; moving agents and spreading fires should have shorter time-to-live.

Relay should therefore test burst loss, not just iid packet drop, and messages should carry age/TTL semantics. A receiver can fall back to the last known message but must know when to stop trusting it.

A delta-triggered sender is also a simple strong baseline for sparse communication before implementing learned prediction-based scheduling.

Relay experiments inspired by this paper

- iid packet drop vs burst/Markov-style loss at matched average loss rate.
- Last-message cache with multiple TTL values.
- Static-information TTL vs dynamic-information TTL.
- Send-on-change threshold vs learned communication gate.
- Measure catastrophic failures caused by stale cached fire/agent state.

Bottom line: Essential for realistic loss experiments and for exploiting temporal redundancy without pretending old information stays valid forever.

15. Communication Learning via Backpropagation in Discrete Channels with Unknown Noise

Authors: Benjamin Freed, Guillaume Sartoretti, Jiaheng Hu, Howie Choset

Year / venue: 2020 - AAAI 2020

Primary link: <https://ojs.aaai.org/index.php/AAAI/article/view/6205>

Base / research question

Continuous differentiable communication is easy to train but unrealistic as a literal digital radio protocol. This paper asks how agents can learn through a discrete channel whose corruption/noise process is not known well enough to differentiate through directly.

Its significance is methodological: it tries to retain end-to-end gradient learning while respecting a discrete communication interface and unknown channel noise.

Approach / method

The authors construct stochastic encoder/decoder transformations that make the discrete noisy channel amenable to gradient-based learning. An additional transformation is used so that channel noise can be handled without requiring the policy to know an exact differentiable noise model.

The approach is evaluated on multi-robot path-finding and collaborative-search tasks, comparing discrete-channel learning with real-valued differentiable communication.

Key findings / conclusions

The paper shows that agents can learn useful cooperative communication over discrete noisy channels with learning speed and performance that remain competitive with continuous differentiable communication in the studied tasks.

The practical conclusion is that discrete execution does not necessarily require abandoning end-to-end learned communication. The training interface can be designed to bridge the gradient/discreteness mismatch.

Gaps / limitations

The mathematical construction is specialized and predates several simpler modern gradient estimators/quantization tricks that may be easier to integrate with MAPPO.

The work focuses on channel differentiability/noise rather than modern questions such as learned recipient selection, variable message size or long latency.

The exact assumptions required by the stochastic transformations should be checked carefully before claiming direct applicability to an arbitrary wireless model.

Implications for Relay

Relay should begin with continuous communication because it is the cleanest way to debug emergent coordination, but the final project should not stop there if it makes claims about constrained communication. This paper provides a principled route toward discrete execution.

A practical Relay study could compare straight-through/Gumbel-style discrete symbols against this class of stochastic-channel approach under bit flips or symbol corruption. The purpose is not necessarily to reproduce the entire method, but to verify that the protocol survives genuine discrete/noisy transmission.

This closes the literature chain: representation bottleneck -> discrete vocabulary -> channel corruption -> task-level robustness.

Relay experiments inspired by this paper

- Continuous communication vs discrete symbol communication.
- Vocabulary/bit-width sweep under equal task conditions.
- Bit/symbol flip probability sweep.
- Train clean, test noisy vs train with noise.
- Compare simple straight-through/Gumbel estimator with a noise-aware discrete-channel method.

Bottom line: Key bridge from convenient differentiable messages to genuinely discrete, noisy communication.

Cross-paper conclusions for Relay

Do not conflate message dimension, entropy and bits. Use separate terminology and metrics. A 4-D float vector is a representation constraint; a 4-symbol categorical message is a discrete protocol; entropy estimates the average information content; physical bandwidth adds channel-level overhead.

Build a clean baseline before adding communication sophistication. Start with MAPPO + continuous messages in a deterministic clean environment. If the base task does not learn, latency, quantization and gating will only make debugging harder.

Evaluate the communication policy, not only task reward. Log who sent, who received, message size, message age, message entropy/code, gating decisions and semantic events. Without communication logs, later protocol analysis will be weak.

Use role structure as a scientific variable. Compare shared versus role-specific communication modules, targeted versus broadcast routing, and changed team compositions. Relay has a natural advantage here because role heterogeneity is part of the task design rather than an artificial benchmark add-on.

Separate channel impairments before combining them. Packet drop, burst loss, symbol corruption and latency create different failure modes. Characterize each independently, then evaluate compound stress tests.

Prefer Pareto curves to one "best" communication setting. Report success/return against messages per episode, symbols/bits per episode, latency or loss. The useful result is often a tradeoff frontier rather than a single scalar score.

Interpretability should be empirical, not assumed. Probe whether message clusters correlate with victim discovery, fire discovery, coordinates, role identity, urgency or route information. Do not label an emergent symbol with human semantics unless intervention/ablation supports the claim.

Suggested deep-read priority

Tier A - directly shapes Relay experiments: HetNet; VIL2C; BVME; CoDe; Effective Multi-Agent Communication Under Limited Bandwidth; IMAC; ETCNet; TMC.

Tier B - strong module/analysis ideas: Sparse Communication/MBC; T2MAC; DSMS; Communication Learning via Backpropagation in Discrete Channels with Unknown Noise.

Tier C - advanced receiver-side refinements: M2I2; DRMAC; COCOM.

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